

AgriScan AI: Crop Disease Detection Using CNN

Comprehensive Academic Project Report & Defense Viva Guide

Author: **Virat Dagar** | ABES Engineering College (CSE Data Science)

Program: HCLTech Industry Aligned Projects (Academic Session 2026–2027)

1. Executive Summary & Product Vision

Crop diseases pose severe threats to global agricultural yield and food security. Early, accurate identification enables targeted intervention, minimizing chemical usage and crop loss. **AgriScan AI** is a multi-tier computer vision decision-support system built using Deep Learning and Convolutional Neural Networks (CNN). The system incorporates **Approach 1: Binary Leaf Classifier (Gate Model)** as a primary Stage 1 security gatekeeper, followed by a 38-class Stage 2 fine-grained crop disease classifier. End-to-end web deployment is handled via a lightweight Flask web interface.

Parameter	Details
Project Title	Crop Disease Detection Using Convolutional Neural Networks (CNN)
Student Author	Virat Dagar
Department & College	Computer Science & Engineering (Data Science), ABES Engineering College
Industry Alignment	HCLTech Industry Aligned Projects Program
Stage 1 Architecture	Approach 1: Binary Leaf Classifier (Gate Model - Is Leaf vs Non-Leaf)
Stage 2 Architecture	Baseline 4-Block Conv2D Neural Network (38 Target Classes)
Dataset Scale	PlantVillage Dataset (54,306 images across 14 crops & 38 classes)
Backend Web Framework	Flask 3.0 (Python 3.12)

2. System Architecture & Methodology

The system operates on a **Hierarchical Multi-Stage Pipeline** to prevent invalid non-leaf images from producing misleading disease predictions:

- Stage 1 (Gate Model - Approach 1):** The uploaded image passes through a dedicated Binary CNN Gate Model (3 Conv2D blocks + Sigmoid). If the model determines the image is not a valid plant leaf (Probability < 50%), the pipeline rejects the input immediately with actionable feedback.
- Stage 2 (Disease Classifier):** Valid leaf images are resized to 128x128x3 and normalized to [0, 1]. A 4-block Conv2D network computes Softmax probabilities across 38 crop disease classes.
- Stage 3 (Agronomic Advisory):** The predicted label maps to disease_info.json, providing symptoms, cause, and organic/chemical management guidelines.

3. Convolutional Neural Network Layer Specifications

Layer Type	Kernel / Units	Activation	Purpose
Input	(128, 128, 3)	None	Raw RGB image tensor normalized to [0,1]
Conv2D Block 1	32 filters (3x3)	ReLU	Extracts low-level edges, colors, and line features
Conv2D Block 2	64 filters (3x3)	ReLU	Extracts textures, leaf veins, and spot boundaries
Conv2D Block 3	128 filters (3x3)	ReLU	Identifies pathological lesions, rusts, and fungal specks
Conv2D Block 4	256 filters (3x3)	ReLU	Encodes high-level spatial disease representations
Dense Layer	256 units	ReLU	Fully-connected decision feature synthesis
Output Layer	38 units	Softmax	Computes categorical probability distribution across classes

4. Teacher Viva Voce Defense & Examination Guide

Study these core technical questions and answers for your project evaluation:

Q1: What is Approach 1 (Binary Leaf Classifier / Gate Model) and why is it essential?

Answer: Approach 1 acts as a Stage 1 security gatekeeper in the hierarchical architecture. It uses a binary CNN to answer 'Is this image a plant leaf?' (Class 1 = Leaf, Class 0 = Non-Leaf). Without this gate model, uploading a picture of a car or shoe would force a standard 38-class classifier to output an arbitrary crop disease prediction with misleading confidence.

Q2: Why use CNNs instead of traditional ML algorithms like SVM or Random Forest?

Answer: Traditional ML requires manual, handcrafted feature extraction (e.g., SIFT, color histograms) which cannot capture complex leaf disease textures across 38 classes. CNNs automatically learn hierarchical visual features directly from pixel matrices (Edges -> Textures -> Lesions -> Disease Patterns) with spatial and translation invariance.

Q3: What loss function did you use for multi-class classification and why?

Answer: SparseCategoricalCrossentropy. Unlike standard CategoricalCrossentropy which requires memory-intensive one-hot encoded vectors for 38 classes across 54,000 images, SparseCategoricalCrossentropy accepts integer labels directly, achieving identical mathematical loss computation with significantly lower memory overhead.

Q4: How do Batch Normalization and Dropout prevent overfitting?

Answer: Batch Normalization normalizes layer outputs (mean=0, variance=1) across mini-batches, accelerating convergence and stabilizing training. Dropout randomly deactivates a fraction of neurons (e.g., 25% to 50%) during forward passes, preventing co-adaptation of features and forcing the network to learn redundant representations.

Q5: What metrics were used to evaluate model performance beyond overall accuracy?

Answer: We evaluated Precision ($TP / (TP + FP)$) to minimize false positives (avoid misdiagnosing healthy crops), Recall ($TP / (TP + FN)$) to minimize missed disease infections, F1-Score (harmonic mean balancing precision and recall), and a 38x38 Confusion Matrix to identify inter-class confusion.

Q6: What are the limitations of this model in real-world agricultural settings?

Answer: PlantVillage images were captured under uniform laboratory lighting. Real-world field leaves contain complex soil/foilage backgrounds and varying sunlight. To address this, Approach 1 (Binary Gate) filters bad background inputs, while data augmentation improves generalization.

Q7: How is the web application deployed and structured?

Answer: The application uses Flask backend (app.py) serving an HTML5/CSS3/JS user interface. Images uploaded via REST API (/predict) are validated, passed to the Stage 1 Gate Model, processed by the Stage 2 CNN, and returned with JSON diagnostic results and management advice.

5. Conclusion & Academic Summary

AgriScan AI demonstrates an end-to-end industrial deep learning application. By combining **Approach 1: Binary Leaf Classifier (Gate Model)** with a 38-class baseline CNN and a Flask decision-support web interface, the project achieves robust, reliable agricultural disease identification.